

Storyboard Template

Completed Storyboard – Build-Ready · Revision 2

Revision Notes · v1.1

This revision resolves three review comments on v1.0, all in Gameplay 2, 3, and the Ending. Each is addressed directly in the affected node table below, and summarised here for quick reference.

1. Which vulnerable question gets redesigned?

v1.0 silently carried only the market-segmentation question into Gameplay 2, with no stated reason for dropping the noble-gases question. Resolved: Gameplay 2 now redesigns both flagged vulnerable questions in sequence. Nothing found vulnerable in Gameplay 1 is silently discarded. See Node 03.

2. What happens on a misclassification in Gameplay 3?

v1.0 did not state what the system does if the learner misjudges the cold-application question. Resolved: Gameplay 3 is now explicitly specified as a silent-log scene – no mid-scene correction, no forced retry, and no reveal of correctness. The result is recorded as given and explained causally at the Ending. This is stated as a deliberate design choice, not an oversight: correcting the learner mid-test would defeat the purpose of an unaided check. See Node 04.

3. How does a learner get more practice after Gameplay 3?

v1.0 offered no path to try a different unaided question, which read as a dead end for anyone who wanted another attempt. Resolved: the Ending now offers an optional, non-gating ‘Practice Again’ control that loads a new cold-application question from a small bank and returns to the Ending afterwards. This does not add a sixth top-level scene – it is a re-entry into Gameplay 3 – and it does not overwrite the first, official, unaided result used for scoring. See Node 04 and Node 05.

1 · Cover & Metadata

Course / topic	Module 2 – AI Literacy for Teaching and Research · Unit 2.4. Competency area: spotting a weak or AI-vulnerable assessment question using the flip test, and redesigning it to be course-specific. Companion to the Unit 2.4 video, AI for Assessment and Feedback, and the MCQ-Generation and Integrity-Resilient Patterns interactive PDF.
Simulation title	Spot the Weak Question – The Flip-Test Challenge
Designer / Builders	[Project team / builder name to be inserted]
Date / version	20 September 2026 · Storyboard v1.1 (revised following review)
Target learners	EKSU academic staff completing AI Literacy for Teaching and Research, Unit 2.4, immediately after meeting the flip test in the Unit 2.4 video. Treat as capable, senior professionals practising a habit they have just watched demonstrated, not as beginners meeting the idea for the first time.
Estimated duration	TBC through prototype testing. Untimed throughout – this simulation rewards a sound judgement, not a fast one.
Setting	A small stack of exam questions sitting on the learner's desk, exactly like the ones flip-tested in the Unit 2.4 video – one of them is the same market-segmentation question the video already tested.

2 · Simulation Type Selector

Tick and combine as needed (see Instructions Guide, Section 3, for full descriptions):

- ☐ Dialogue / branching
- ☐ Evidence-gathering / environment-emergent exploration
- ☒ Instrumented testing – the flip test itself: each question is sent to a simulated AI and the reply is observed before judgement
- ☐ Procedural sequencing

- ☐ Token / resource allocation
- ☐ Drag-and-drop / annotation builder – not required; select-and-confirm and a single editable text field are used instead
- ☒ Others: vulnerable/resilient classification; redesign-and-retest loop applied to every flagged question; one unaided cold application with an optional repeat from a small question bank

Tone, voice & reading level

Warm, plain and respectful of an experienced professional audience. No jargon beyond what the Unit 2.4 video already introduced (the flip test, course-specific detail). British English throughout. Short, direct on-screen copy; no re-teaching of the flip test during play – this simulation assumes the video, not the other way around.

Simulation visual direction: reuse the established Digi-Teach simulation system – warm cream base, teal for headers and structural elements, amber and sky as supporting accents, orange reserved for interactive callouts. Baloo 2 for headings and Nunito for body/UI text.

Do not use colour, icon choice or position to disclose whether a question is vulnerable or resilient before the learner runs the flip test themselves.

4 • The Premise

The setting / world

The learner has a short stack of exam questions to check before they go anywhere near students. One is the exact market-segmentation question already flip-tested in the Unit 2.4 video. The rest are new, but the same shape of problem.

The experience begins with the questions themselves, not with a restatement of what the flip test is or why it matters.

The learner's role

The learner runs the flip test on each question, judges whether it is vulnerable or resilient from what the test shows, and redesigns every question flagged vulnerable using course-specific detail – then proves the habit survives on a question they have not seen before.

The problem / situation

Some of these questions can be answered instantly by AI with no knowledge of the course at all. Others cannot, because they depend on something only this class actually covered. The learner must tell the difference using the flip test, not guesswork or gut feel.

The goal / objective

Correctly flip-test and classify the starting questions, redesign every question flagged vulnerable until each genuinely resists the test, and apply the same routine, unaided, to a question from a different subject – directly extending the habit the Unit 2.4 video introduced.

5 · State & Scoring Framework

ACCURACY / STATE STRANDS	SUMMARY
Strand 1 – Flip-Test Accuracy	Measures whether the learner correctly classifies each starting question as Vulnerable or Resilient after running the flip test. Updated in Gameplay 1.
Strand 2 – Redesign Effectiveness	Measures whether the learner's redesigned version of each flagged question genuinely resists the flip test on retest, using course-specific detail rather than cosmetic rewording. Both flagged questions from Gameplay 1 are redesigned, and both are scored – this strand is never based on a single, silently chosen question. Updated in Gameplay 2.
Strand 3 – Cold Application	Measures whether the learner can correctly flip-test, classify and propose a course-specific fix for a new, unfamiliar question with no walkthrough or hints. Scored from the learner's first, official attempt only in Gameplay 3; any optional additional attempts made via the Ending's 'Practice Again' control are recorded as extra practice evidence and never overwrite or improve this score. Updated in Gameplay 3.

Running log (if applicable)

A quiet decision trace stores: each Gameplay 1 flip-test result and classification; each Gameplay 2 redesign question's text, retest results and final status; the Gameplay 3 official question, classification, proposed fix and result; and, if used, every optional additional Gameplay 3 attempt made from the Ending. It supports the ending audit and is not presented as a live points display during play.

6 · Structure & Flow

Use one title / intro, three major gameplay sequences and one ending / conclusion. Internal panels and reveal states are not additional top-level scenes. The optional practice loop described in Node 04 and Node 05 re-enters Gameplay 3 with new content; it does not create a sixth scene.

STANDARD MIVA SHELL

MIVA UNIVERSITY OPENING	OPTIONAL INTRO SEQUENCE	TITLE / LANDING PAGE / Student Name	SIMULATION BEATS 1-5
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The standard shell does not count as additional simulation gameplay. The optional pre-simulation intro is not required. Scene 1 performs the formal Title / Intro function within the five-beat simulation architecture.

Scene 1 – The Question Stack

NODE 01 – INTRO · The Question Stack	
Situation	The learner sees three exam questions sitting in a short stack, waiting to be checked before use. One is immediately familiar: ‘Define market segmentation.’ – the same question already flip-tested in the Unit 2.4 video. Required on-screen copy: <i>“Before any of these go in front of a student, run the test you saw in the video. Ready?”</i> One primary control: Start the Audit.
Interaction	Click, tap or keyboard-activate Start the Audit. This is orientation only – no assessed decision is made here.

Response / indicator	The framing recedes. The question stack becomes active, each with an Ask AI control. No score or correctness message is shown yet.
Teaching content	No explicit teaching content is delivered here. The scene deliberately reuses one question the learner already watched tested, so they enter the simulation recognising the routine rather than meeting it cold.
State change	Initialise simulation session. Set: flip_test_accuracy = not started; redesign_result = pending; cold_application = pending.
Goes to	→ Gameplay 1 – Run the Flip Test

Scene 2 – Run the Flip Test

NODE 02 – GAMEPLAY 1 · Run the Flip Test	
Situation	Three questions sit in the stack, each with its own Ask AI control: “Define market segmentation.” “List the noble gases.” “Using the primary source document we analysed in week six, explain why the author’s account differs from the textbook version.”
Interaction	For each question, the learner presses Ask AI to see a simulated reply, then classifies the question as Vulnerable or Resilient based on what the reply shows. The classification control for a question is disabled until Ask AI has been run for that question at least once, so a learner cannot classify from a guess alone. All three questions must be tested and classified before the scene ends.
Response / indicator	“ Define market segmentation. ” – AI replies instantly and completely: ‘Market segmentation is the process of dividing a market into distinct groups of buyers with different needs or behaviours.’

	“List the noble gases.” – AI replies instantly and completely: ‘The noble gases are helium, neon, argon, krypton, xenon and radon.’ “Using the primary source document we analysed in week six, explain why the author’s account differs from the textbook version.” – AI cannot answer: ‘I don’t have access to the specific primary source your class analysed in week six, so I can’t compare it to the textbook account.’
Teaching content	Reinforces the video’s own routine – ask AI to answer cold, then judge from what happens – by having the learner run it themselves rather than watching it run. Confirms that instant, complete, context-free answers are the warning sign, and that a failed or vague AI reply is the sign of a resilient question.
State change	flip_test_result and classification stored per question (correct / incorrect against Vulnerable, Vulnerable, Resilient). Both questions classified Vulnerable – the market-segmentation question and the noble-gases question – are flagged and carry forward into Gameplay 2. Nothing flagged vulnerable in this scene is dropped or resolved automatically; Gameplay 2 addresses every flagged question in turn.
Goes to	→ Gameplay 2 – Redesign and Retest

Scene 3 – Redesign and Retest

NODE 03 – GAMEPLAY 2 · Redesign and Retest	
Situation	Both questions flagged Vulnerable in Gameplay 1 are addressed here, one at a time, in the order they were flagged: first the market-segmentation question, then the noble-gases question. Nothing is carried forward silently and nothing is discarded – the learner works through the full flagged set.

	For each question in turn, an editable text field is seeded with its original wording. Beside it, the same Ask AI control and reply panel from Gameplay 1.
Interaction	For the question currently in focus, the learner edits it to add course-specific detail – a case study, a dataset, or something discussed in a named session – then presses Ask AI to retest. They may revise and retest up to two times per question. Once a question reaches Resilient, or the two-attempt cap is reached, the scene automatically advances to the next flagged question; reaching the cap without success does not block progress, and the unresolved status is carried forward honestly rather than hidden.
Response / indicator	Market-segmentation question: A cosmetic rewording with no course-specific detail still returns an instant, complete AI reply – still Vulnerable. A genuine course-specific redesign, such as <i>‘Using the case study we discussed in week four, which segmentation variable explains their targeting strategy, and why?’</i>, returns: ‘I don’t have the details of the case study discussed in your week four session, so I can only guess generally at segmentation variables without more context.’ – now Resilient. Noble-gases question: A cosmetic rewording such as ‘Name the noble gases in order’ still returns an instant, complete AI reply – still Vulnerable. A genuine course-specific redesign, such as <i>‘Using the reactivity data we recorded in the week-three lab, rank these three noble gases and explain the pattern you observed.’</i>, returns: ‘I don’t have the reactivity data recorded in your week-three lab, so I can’t complete this ranking with any confidence.’ – now Resilient.
Teaching content	Makes the video’s redesign move experiential, across more than one example: adding a named, specific, course-only reference – not just rephrasing – is what actually changes the AI’s ability to answer, whether the underlying question is about a case study or a

	lab result. Working through both flagged questions also shows that the fix is a repeatable move, not a trick that only worked once.
State change	For each flagged question: every retest attempt's text and result are logged, and redesign_result stores the final version and whether it reached Resilient, capped at two attempts. Strand 2 is scored across both questions, not one.
Goes to	→ Gameplay 3 – One More, Unaided

Scene 4 – One More, Unaided

NODE 04 – GAMEPLAY 3 · One More, Unaided	
Situation	A new question from a different subject appears: ‘Explain the water cycle.’ No question stack, no worked example alongside it – just the question, the Ask AI control, and a blank field for a proposed fix. This is the learner's official, unaided attempt.
Interaction	The learner presses Ask AI once, classifies the question as Vulnerable or Resilient, and – if they judge it Vulnerable – types one short course-specific detail they would add to strengthen it. There is no second attempt on this specific question within the scene itself.
Response / indicator	AI replies instantly and completely with a standard explanation of the water cycle – Vulnerable. The learner's typed fix is accepted as written; no correctness judgement is applied to exact wording, only whether it names something specific to a real class rather than a generic instruction like ‘add more detail’. On misclassification: if the learner classifies this question as Resilient, the simulation does not interrupt, correct, or force a retry in the moment. There is no on-screen right/wrong signal here. The response is accepted and logged exactly as given, and the correct classification, with its reasoning, is revealed and explained only in the Ending debrief. This is a deliberate design choice: correcting the learner mid-test, or forcing an immediate retry, would test whether

	they can be led to the right answer rather than whether the habit has genuinely transferred unaided.
Teaching content	Tests whether the flip-test habit transfers to an unfamiliar subject without a worked example alongside it, which is the actual target of this unit – not repeated success on a question the learner has already seen handled.
State change	cold_application_classification and proposed_fix text are stored as the official Gameplay 3 result, used for Strand 3 scoring. This result is never overwritten by a later practice attempt (see Node 05).
Goes to	→ Ending – Your Question Audit

Scene 5 – Your Question Audit

NODE 05 – ENDING / CONCLUSION · Your Question Audit	
Situation	A restrained audit screen, not a congratulations panel. It shows: the three Gameplay 1 classifications, both Gameplay 2 redesigned questions with their final results, and the official Gameplay 3 question with the learner's classification and proposed fix.
Interaction	Review only for the core simulation. A Finish / Continue action closes the simulation and returns to the programme. Optional practice: a secondary control, 'Practice Again', is also available here. Selecting it loads one further cold-application question, drawn at random without repeats from a small bank of two additional questions, and re-enters Node 04 – Gameplay 3 with that new question. Completing it returns the learner to this same Ending screen, now showing the additional attempt alongside the original. This loop may be repeated until the small bank is exhausted. It is entirely optional and never required to finish the simulation.
Response / indicator	Results are shown plainly – three classifications, two redesign outcomes, one official unaided result, and any additional practice attempts. No confetti, badges or pass/fail label. If the official

	cold-application question was misclassified, the gap is named plainly rather than softened, and the correct classification and reasoning are given here for the first time.
Teaching content	The debrief reconstructs the chain: an instant, complete, context-free AI reply is the warning sign; a genuine fix names something only this class actually has; and the real test of the habit is whether it survives on a question with no example running alongside it. If the official attempt was misjudged, the debrief explains what the AI's reply actually showed and why, so the correction is causal rather than a bare right/wrong flag. Closing transfer line: <i>“This is exactly what your unit task asks for next: take one real question from your own course, and redesign it the same way. If you'd like more practice first, use Practice Again above – it won't change the result already recorded here.”</i>
State change	Finalise flip_test_accuracy, redesign_result (both questions) and the official cold_application result. Any optional practice attempts are appended to the session record as additional evidence, clearly separated from the official Strand 3 result. No additional gameplay state beyond this is introduced.
Goes to	→ Simulation close / return to programme

Practice question bank (for the optional ‘Practice Again’ loop)

Two additional cold-application questions, held in reserve for learners who choose to practise further after their official Gameplay 3 attempt. Neither is used anywhere else in the simulation, and neither affects the official Strand 3 score.

Bank question A: “Explain how vaccines create immunity in the body.”

Vulnerable. AI replies instantly and completely with a standard, textbook explanation of how vaccines work – no course context needed.

Bank question B: “Using the dataset we reviewed in week nine, which vaccine trial showed the strongest results, and why?”

Resilient. AI cannot answer: it has no access to the specific week-nine dataset, so it can only speak in generalities.

Both are included deliberately, not only a further vulnerable example, so the practice loop keeps testing genuine classification judgement rather than letting the learner assume every cold question will turn out to be vulnerable.

7 · Audio/Video Production Spec (if applicable)

Format & duration	UI-native interactive simulation. No required video. Untimed throughout. Duration TBC through prototype testing.
Voices required	None required. The learning depends on running a test and judging its result, not on performed dialogue.
Visual style	Premium, uncluttered 2D interface, matching the established Digi-Teach simulation system: warm cream background, teal structural headers, amber and sky accents, orange for interactive callouts. Baloo 2 headings, Nunito body/UI. One question and its test result in focus at a time.
Accessibility	No essential information depends on sound. If optional sound is added later it remains non-instructional. Any future voice content requires default-on captions and transcript.
Fixed vs. open	Fixed: five-beat 1+3+1 structure; the three starting questions and their flip-test results; the redesign-and-retest logic applied to both flagged questions from Gameplay 1; the single-official-attempt cold application in Gameplay 3, with its result locked before any optional practice; the two-question practice bank and its non-scoring status; the ending audit comparison. Open: transition animation, exact simulated-reply wording beyond the required examples, microcopy tone, subtle sound design.

Full transcript

No voiced transcript is required. Required core on-screen lines:

Intro: *“Before any of these go in front of a student, run the test you saw in the video. Ready?”*

Primary action: *“Start the Audit”*

Gameplay 1 prompt: *“Ask AI. Then decide: vulnerable, or resilient?”*

Gameplay 2 prompt: *“Add something only this class has. Then test it again.”*

Gameplay 3 prompt: *“One more – no example this time. Test it, judge it, and name your fix.”*

Ending practice prompt: *“Want another go? Practice Again loads a new question – it won't change the result already recorded.”*

Closing transfer: *“This is exactly what your unit task asks for next: take one real question from your own course, and redesign it the same way.”*

8 · Debrief & Leaderboard

How the tracked strands combine into a final result

No weighted composite score and no pass/fail label are created. The learner's own flip-test classifications, redesign outcomes and cold-application result are the headline outcome – reported as three strands, not combined into one number:

- Flip-Test Accuracy – how many of the three starting questions were correctly classified in Gameplay 1.
- Redesign Effectiveness – whether each of the two flagged questions (market segmentation and noble gases) reached Resilient, and in how many attempts.
- Cold Application – the classification and proposed fix for the official, first-attempt Gameplay 3 question. Any additional practice attempts from the Ending are shown separately and do not alter this score.

What's displayed but not scored

- Time spent per question.
- Number of times a simulated reply was re-read.
- Exact wording of intermediate (non-final) redesign attempts.

- Any optional Practice Again attempts and their outcomes, shown for the learner's own reference only.

Leaderboard

None. The learning argument is sound professional judgement, not competition between learners. Do not rank learners by accuracy or speed.

Backend

Implementation technology is not prescribed here. Minimum tracked fields: learner/session identifier; Gameplay 1 per-question classification; Gameplay 2 redesign attempts and final result for both flagged questions; Gameplay 3 official classification and proposed fix; any optional practice attempts, clearly flagged as non-scoring; completion status. A facilitator dashboard, if required by programme analytics, should expose the raw strand evidence rather than a fabricated single performance score.

9 · Accessibility Requirements

- ☐ Default-on subtitles for all voiced content, minimum 18px, high contrast – not applicable to the current design because no voiced content is required. If voice is added later, this requirement becomes mandatory.
- ☒ Keyboard or tap/click alternative for every selection, the Ask AI control and submitting a redesign or fix – no drag is required anywhere in this simulation.
- ☐ Text-label equivalents for directional / compass navigation – not applicable; no directional navigation is used.
- ☒ Accessible labelling for the Ask AI control, the Vulnerable/Resilient classification buttons, and the redesign and fix text fields.
- ☒ Minimum 44px touch targets throughout.

Additional simulation-specific requirements:

- Every question card, the Ask AI control, and both classification buttons must be keyboard reachable in a logical focus order.
- Enter / Space must support triggering Ask AI and confirming a classification.
- The redesign and proposed-fix text fields require explicit visible labels, not placeholder-only text.

- Simulated reply outcomes (instant/complete vs. unable to answer) must be conveyed with a text label in addition to any visual treatment, never colour alone.
- The remaining-attempts count in Gameplay 2 must be programmatically available to assistive technology, for whichever of the two flagged questions is currently in focus.
- The Ending's Practice Again control must be labelled clearly as optional and non-scoring, both visually and to assistive technology, so no learner mistakes it for a required step.
- No accessibility route may reveal the correct classification or the redesign answer earlier than the standard route.
- Animations and transitions must respect reduced-motion preferences.
- Interface contrast must remain sufficient across the cream, teal, amber, sky and orange palette.

Equivalent routes preserve the same information, agency, difficulty, consequence and assessment validity.

Structural Acceptance Summary

The simulation contains exactly five primary beats:

1. Title / Intro – The Question Stack
2. Gameplay 1 – Run the Flip Test
3. Gameplay 2 – Redesign and Retest (both flagged questions)
4. Gameplay 3 – One More, Unaided (with an optional practice loop, re-entering this same scene, offered only from the Ending)
5. Ending / Conclusion – Your Question Audit

The market-segmentation question and its redesign are the same example already shown in the Unit 2.4 video, so the learner enters Gameplay 1 and 2 recognising the routine rather than meeting it cold. No new testing method, term or concept is introduced here – only guided, then unaided, practice running the flip test the video taught.

This revision closes three open questions raised in review: every question flagged vulnerable in Gameplay 1 is redesigned in Gameplay 2, not only one; Gameplay 3 explicitly logs a misclassification silently and explains it only at the Ending, by design; and the Ending's optional Practice Again loop gives a learner who wants it a way to attempt further

unaided questions without disturbing the official, scored result or adding a sixth top-level scene.